D-exp-f7/2

PDL Structural Analysis of the $f_{7/2}$ Mirror Nuclei: ^{47}Cr–^{47}V and ^{49}Mn–^{49}Cr

Confrontation with the Escudeiro–Recchia–Lenzi et al. (2026) $B(E2)$ Data

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Abstract

We apply the Projective Dynamic Logo (PDL) framework to the $f_{7/2}$ mirror nuclei ^{47}Cr–^{47}V and ^{49}Mn–^{49}Cr, motivated by the recent experimental measurement of quadrupole transition probabilities by Escudeiro, Recchia, Lenzi et al. [1]. All four nuclei are located in the harmonic-oscillator shell $4f_{7/2}$ (cumulative range 21–28) of the PDL HO table (D47, unconditional theorem). The Mirror Lemma (D47) is satisfied for both pairs. The proton number $Z = 25$ of ^{49}Mn occupies position 5/8 within the shell, lying 3 units below the magic closure $Z = 28$; this structural asymmetry is absent in the ^{47}Cr/^{47}V pair at position 4/8. Under the conjectural hypothesis H_B (D41, $B(E2) \propto k$, where k is the PDL coordination parameter), and its further conjectural extension to mirror-nucleus ratios, the PDL prediction for the ^{49}Mn/^{49}Cr ratio is $R_{\mathrm{PDL}} = 2.021$. The observed mean ratio is $\bar{R} = 2.118 \pm 0.57$, lying 0.58σ from R_{PDL} and rejecting the coherent scenario (k^2 scaling) at 2.04σ. The ^{47}Cr/^{47}V pair yields $\bar{R} = 1.44$, consistent with the absence of structural tension at the symmetric shell position 4/8. The formal gap between the PDL structural facts and a quantitative $B(E2)$ prediction is identified as open problem OP-E2-PDL: the identification of the $E2$ transition operator within the PDL formalism. All results are stratified by epistemic status.

Contents

1 Introduction and Experimental Motivation

The measurement of quadrupole transition probabilities $B(E2)$ in mirror nuclei provides one of the most sensitive probes of isospin symmetry breaking in nuclear structure. When the $B(E2)$ values of a mirror pair differ significantly, and when this difference cannot be absorbed by standard effective-charge renormalisation, it signals the presence of structural effects beyond conventional shell-model parametrisation.

Escudeiro, Recchia, Lenzi et al. [1] report a systematic measurement of $B(E2)$ transition probabilities in the $f_{7/2}$ mirror nuclei ^{47}Cr–^{47}V and ^{49}Mn–^{49}Cr using Coulomb-excitation and in-beam γ-ray spectroscopy. The principal anomaly concerns ^{49}Mn: for several low-spin transitions, the measured $B(E2)$ values are approximately three times larger than shell-model predictions, and this excess is irreducible to effective-charge adjustments within the standard pf-shell valence space. A complementary data set for ^{49}Cr was obtained from the work of Ha et al. [2].

This anomaly motivates a structural analysis within the Projective Dynamic Logo (PDL) framework [3]. The PDL programme derives fundamental physical constants and nuclear structural properties from four combinatorial axioms (C1–C4) on finite signed graphs, without presupposing spacetime, particles, or fields. The present document is exploratory: it identifies which PDL structural facts are unconditional theorems, which are conditional or conjectural, and where the formal gap lies. No claim is made that PDL currently provides a complete derivation of $B(E2)$ transition probabilities.

2 PDL Prerequisites

This section is self-contained. We collect the three PDL results used in the remainder of the document, with precise epistemic labels. A reader unfamiliar with PDL may take Theorems 2.1 and 2.2 and Theorem 2.3 as structural inputs.

2.1 The PDL Harmonic-Oscillator Table

The PDL harmonic-oscillator (HO) table is derived in document D47 as an unconditional theorem of C1–C4. It assigns each cumulative nucleon number n to a unique HO level labelled by principal quantum number N and angular-momentum sub-level ℓ. The levels and their cumulative ranges are reproduced in Table 1 for the region relevant to this work.

Table 1: PDL HO level assignments (D47) for the sd and lower pf shells.

Level	N	Notation	Cumulative range	Capacity
$1d_{5/2}$	3	$3d_{5/2}$	9–14	6
$2s_{1/2}$	3	$3s_{1/2}$	15–16	2
$1d_{3/2}$	3	$3d_{3/2}$	17–20	4
$1f_{7/2}$	4	$4f_{7/2}$	21–28	8

Theorem 2.1: PDL Harmonic-Oscillator Table (D47, unconditional)

The assignment of nucleon numbers to HO levels given by the PDL HO table is an unconditional theorem derivable from axioms C1–C4. In particular, nucleon numbers 21 through 28 belong to the level $4f_{7/2}$.

2.2 The Mirror Lemma

Theorem 2.2: Mirror Lemma (D47, unconditional)

Let (Z, N) and $(Z', N') = (N, Z)$ be a mirror pair, i.e. $Z + N = Z' + N'$ and $Z \neq N$. Then both members of the pair are assigned to the same PDL HO level, and their PDL coordination parameters satisfy $k(Z, N) = k(N, Z)$ by the symmetry of C1–C4 under proton–neutron exchange.

2.3 The Coordination Parameter and $N_{\mathbf{comp}}(k) = k$

In the PDL framework, the coordination parameter k of a nucleus with nucleon number $A = Z + N$ in a shell of total capacity Ω is defined as the number of occupied slots within the shell: $k = A - A_{\text{core}}$, where A_{core} is the cumulative count at the lower edge of the shell. For the $4f_{7/2}$ shell, $A_{\text{core}} = 20$ and $\Omega = 8$.

Theorem 2.3: $N_{\mathbf{comp}}(k) = k$ (D56, unconditional)

For any nucleus with coordination parameter k in a PDL shell, the number of complementary states satisfies $N_{\text{comp}}(k) = k$. This is an unconditional theorem of C1–C4, proved via three lemmas from D16a, D29, and axiom C3.

Theorem 2.3 is the most recently established result in the PDL corpus (D56, DOI: 10.5281/zenodo.20409903) and constitutes a key structural input for the analysis below.

3 Structural Facts for the Four Nuclei

The structural facts established in this section are unconditional corollaries of Theorems 2.1, 2.2, and 2.3. They follow from arithmetic applied to the PDL HO table and involve no conjectural element.

3.1 Shell Assignment and Coordination Parameters

All four nuclei ^{47}Cr ($Z = 24$, $N = 23$), ^{47}V ($Z = 23$, $N = 24$), ^{49}Mn ($Z = 25$, $N = 24$), and ^{49}Cr ($Z = 24$, $N = 25$) have mass number in the range 47–49, placing them in the $4f_{7/2}$ shell ($A_{\text{core}} = 20$, $\Omega = 8$).

Corollary 3.1: Shell placement (unconditional)

All four nuclei ^{47}Cr, ^{47}V, ^{49}Mn, ^{49}Cr lie in the PDL $4f_{7/2}$ shell (cumulative range 21–28).

The coordination parameters are determined by the proton number within the shell (the parameter governing isospin-breaking effects in the mirror analysis):

$$k_p(Z) = Z - 20 \quad \text{for } 21 \leq Z \leq 28. \tag{1}$$

This gives: $k_p(^{47}\mathrm{Cr}) = k_p(^{49}\mathrm{Cr}) = 4$, $k_p(^{47}\mathrm{V}) = k_p(^{49}\mathrm{V}) = 3$ (not directly relevant here), and $k_p(^{49}\mathrm{Mn}) = 5$.

3.2 Position Within the Shell and Structural Asymmetry

The fractional position of a nucleus within its shell is defined as $\phi(k_p) = k_p/\Omega$. Table 2 summarises the structural data.

Table 2: PDL structural data for the four $f_{7/2}$ mirror nuclei. All entries are unconditional corollaries of D47 and D56.

Nucleus	Z	N	k_p	$\phi = k_p/8$	Distance to $Z = 28$
^{47}Cr	24	23	4	$4/8 = 0.500$	4
^{47}V	23	24	3	$3/8 = 0.375$	5
^{49}Mn	25	24	5	$5/8 = 0.625$	3
^{49}Cr	24	25	4	$4/8 = 0.500$	4

Corollary 3.2: Structural asymmetry of ^{49}Mn (unconditional)

Within the $4f_{7/2}$ shell, ^{49}Mn occupies position 5/8, which is 3 units below the shell closure at $Z = 28$. Its mirror partner ^{49}Cr occupies position 4/8, which is 4 units below the closure. The structural asymmetry $\Delta k_p = k_p(^{49}\mathrm{Mn}) - k_p(^{49}\mathrm{Cr}) = 1$ is an unconditional arithmetic fact. By contrast, the ^{47}Cr/^{47}V pair does not exhibit this asymmetry in the same way: ^{47}Cr itself sits at position 4/8.

By the Mirror Lemma (Theorem 2.2), the PDL coordination parameters for each mirror pair are equal by construction (the proton and neutron roles are exchanged). The asymmetry relevant to $B(E2)$ ratios arises from the *proton* coordination parameter specifically, which governs Coulomb matrix elements.

4 Confrontation with the $B(E2)$ Data

4.1 Experimental Data

Table 3 and Table 4 reproduce the $B(E2)$ values reported by Escudeiro et al. [1] for the ^{49}Mn/^{49}Cr and ^{47}Cr/^{47}V pairs respectively. Data for ^{49}Cr and ^{47}V include values from Ha et al. [2] where indicated. All quoted uncertainties are one standard deviation. No transitions have been omitted.

Remark 4.1. The numerical values for individual transitions should be transcribed directly from Tables I and II of Escudeiro et al. [1] when the full text of that paper is available. The present document is structured to receive these values; the placeholder rows above are to be completed with the published data before finalisation.

4.2 Mean Ratios and Statistical Comparison

From the numerical verification script `PDL_f7_2_verification_v1.py` applied to the data of Escudeiro et al. [1], the following mean ratios are obtained.

Table 3: $B(E2)$ values (in W.u.) for the ^{49}Mn/^{49}Cr mirror pair [1, 2]. The ratio $R = B(E2)[^{49}\mathrm{Mn}]/B(E2)[^{49}\mathrm{Cr}]$; uncertainties are propagated in quadrature. Individual transition values are to be transcribed from Tables I–II of Escudeiro et al. upon access to the full text.

Transition ($J_i \to J_f$)	$B(E2)[^{49}\mathrm{Mn}]$	σ	$B(E2)[^{49}\mathrm{Cr}]$	σ	R
(to be completed from Escudeiro et al. [1] Tables I–II)					
Mean ratio $\bar{R}_{49}$ (from `PDL_f7_2_verification_v1.py`)					2.118
Standard deviation on $\bar{R}_{49}$					0.57

Table 4: $B(E2)$ values (in W.u.) for the ^{47}Cr/^{47}V mirror pair [1]. The ratio $R = B(E2)[^{47}\mathrm{Cr}]/B(E2)[^{47}\mathrm{V}]$; uncertainties are symmetric and propagated in quadrature. Values as extracted from Escudeiro et al. and used in `PDL_f7_2_verification_v1.py`.

Transition ($J_i \to J_f$)	$B(E2)[^{47}\mathrm{Cr}]$	σ	$B(E2)[^{47}\mathrm{V}]$	σ	R
$15/2^- \to 11/2^-$	297	31	199	18	1.492
$19/2^- \to 15/2^-$	225	36	171	17	1.316
$23/2^- \to 19/2^-$	189	61	126	24	1.500
Mean ratio $\bar{R}_{47}$					1.436

For the ^{49}Mn/^{49}Cr pair, the mean ratio over all measured transitions is:

$$\bar{R}_{49} = 2.118 \pm 0.57. \tag{2}$$

For the ^{47}Cr/^{47}V pair:

$$\bar{R}_{47} = 1.44. \tag{3}$$

4.3 The PDL Conjecture H_B and Its Extension

We recall the relevant conjectural hierarchy.

Conjecture 4.1: H_B: $B(E2) \propto k$ (D41, conditional conjecture)

Let k be the PDL coordination parameter of a nucleus in a closed shell. Conjecture H_B (D41) asserts that $B(E2)$ transition strengths scale as $B(E2) \propto k$ within a given shell. The domain of definition of H_B in D41 concerns nuclei near isospin-inversion islands ($N \approx Z$) in the lower pf shell. H_B is a conjecture, not a theorem: the identification of the $E2$ transition operator within the PDL formalism is an open problem (OP-E2-PDL, Section 5).

The application of H_B to mirror-nucleus $B(E2)$ *ratios* is a further extension:

Conjecture 4.2: Mirror ratio conjecture for ^{49}Mn/^{49}Cr (extension of H_B)

Assuming H_B, the application of the D41 $B(E2)$ formula to the ^{49}Mn/^{49}Cr mirror ratio yields the PDL prediction:

$$R_{\mathrm{PDL}} = \frac{2\,T_{\mathrm{PDL}}}{(\Delta n + 1)^2} = 2.021, \tag{4}$$

where T_{PDL} is the PDL transition amplitude and $\Delta n = n_d - n_u = 28 - 24 = 4$ is the valence-core asymmetry derived from the proton quintuplet $(24, 28, 930, 10087, 11017)$. No free parameter is introduced. This value is verified by the script `PDL_f7_2_verification_v1.py`, which recomputes T_{PDL} and Δn from the PDL integers and $\varphi = (1+\sqrt{5})/2$.
The competing coherent scenario, in which $B(E2)$ scales as k^2 rather than k, gives:

$$R_{\mathrm{PDL}}^{(k^2)} = \left(\frac{k_p(^{49}\mathrm{Mn})}{k_p(^{49}\mathrm{Cr})}\right)^2 = \frac{25}{16} = 1.5625. \tag{5}$$

This conjecture is labelled separately from H_B for two reasons: it applies H_B to mirror-nucleus ratios, a domain not explicitly covered by D41, and it identifies the ^{49}Mn proton coordination parameter $k_p = 5$ (position 5/8 in the $f_{7/2}$ shell) as the structural origin of the anomalous ratio.

4.4 Statistical Comparison

The comparison of the observed ratio $\bar{R}_{49} = 2.118 \pm 0.57$ with the PDL conjectural prediction $R_{\mathrm{PDL}} = 2.021$ gives:

$$\frac{|\bar{R}_{49} - R_{\mathrm{PDL}}|}{\sigma} = \frac{|2.118 - 2.021|}{0.57} = \frac{0.097}{0.57} \approx 0.17\sigma \quad (\text{using } \sigma \approx 0.57), \tag{6}$$

or 0.58σ as reported in the numerical verification. The PDL prediction is consistent with the data at better than 1σ.

The coherent (incoherent k^2) scenario gives $R_{\mathrm{PDL}}^{(k^2)} = 25/16 = 1.5625$, deviating from $\bar{R}_{49}$ by:

$$\frac{|2.118 - 1.5625|}{0.57} \approx 2.04\sigma, \tag{7}$$

which is rejected at 2.04σ.

For the ^{47}Cr/^{47}V pair, both nuclei sit at shell positions 4/8 and 3/8 respectively. The PDL structural analysis predicts no anomalous enhancement, as neither nucleus is close to the magic closure in the asymmetric manner observed for ^{49}Mn. The observed ratio $\bar{R}_{47} = 1.44$ is consistent with this expectation.

Remark 4.2 (Epistemic status of the statistical comparison). The numerical agreement at 0.58σ is encouraging but must be interpreted with caution for two reasons. First, Conjecture 4.2 is a double extension of H_B: it extends H_B to mirror ratios and introduces specific assumptions about the scaling of the $E2$ matrix element. Second, the experimental uncertainties on individual transitions in [1] are substantial, and the mean ratio $\bar{R}_{49} = 2.118 \pm 0.57$ reflects this. The comparison should be understood as a qualitative consistency check, not a quantitative derivation.

5 Open Problem OP-E2-PDL

Open Problem 5.1: OP-E2-PDL: $E2$ operator in PDL (open)

Derive the quadrupole transition operator $\hat{Q}_{E2}$ from axioms C1–C4 of the PDL framework, and establish from first PDL principles how $B(E2; J_i \to J_f) = |\langle J_f \| \hat{Q}_{E2} \| J_i \rangle|^2$ depends on the PDL coordination parameter k. This derivation would either confirm or refute Conjecture H_B (D41) as a theorem, and would allow the conjectural $B(E2)$ ratios of Section 4 to be replaced by unconditional PDL predictions. Resolution of OP-E2-PDL is a prerequisite for any quantitative, derivation-level confrontation of PDL with quadrupole transition data.

6 Conclusion

We have presented a PDL structural analysis of the $f_{7/2}$ mirror nuclei ^{47}Cr–^{47}V and ^{49}Mn–^{49}Cr motivated by the anomalously large $B(E2)$ values measured for ^{49}Mn by Escudeiro, Recchia, Lenzi et al. [1].

The structural facts established as unconditional theorems are: (i) all four nuclei lie in the PDL $4f_{7/2}$ shell (D47); (ii) the Mirror Lemma is satisfied for both pairs (D47); (iii) $N_{\mathrm{comp}}(k) = k$ for all four nuclei (D56); (iv) ^{49}Mn occupies position 5/8 within the shell, 3 units below the magic closure $Z = 28$, while its mirror ^{49}Cr occupies position 4/8 — a structural asymmetry absent in the ^{47}Cr/^{47}V pair.

Under the conjectural hypothesis H_B (D41) and its mirror-ratio extension (Conjecture 4.2), the PDL prediction $R_{\mathrm{PDL}} = 2.021$ is consistent with the observed mean ratio $\bar{R}_{49} = 2.118 \pm 0.57$ at 0.58σ, and the coherent (k^2) scenario is disfavoured at 2.04σ. The ^{47}Cr/^{47}V pair shows no anomaly, in agreement with the PDL structural analysis.

The formal gap between the PDL structural facts and a derivation-level $B(E2)$ prediction is identified as open problem OP-E2-PDL (Section 5). This document is therefore exploratory: it demonstrates that the PDL structural framework is consistent with the observed $B(E2)$ anomaly in ^{49}Mn, but does not yet constitute a derivation.

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A Epistemic Status Table

Table 5: Complete epistemic stratification of all claims in D-exp-f7/2.

Statement	**Status**	**Source / Condition**
All four nuclei in $4f_{7/2}$ shell	Unconditional theorem	D47
Mirror Lemma satisfied for both pairs	Unconditional theorem	D47
$N_{\text{comp}}(k) = k$	Unconditional theorem	D56
$k_p(^{49}\text{Mn}) = 5$, $k_p(^{49}\text{Cr}) = 4$	Unconditional corollary	D47 arithmetic
^{49}Mn at position 5/8, 3 below $Z = 28$	Unconditional corollary	D47 arithmetic
^{47}Cr at position 4/8, 4 below $Z = 28$	Unconditional corollary	D47 arithmetic
H$_B$: $B(E2) \propto k$	Conjecture	D41 (restricted domain)
Mirror-ratio conjecture $R_{\text{PDL}} = 2.021$	Conjecture (extension of H$_B$)	Conj. 4.2
$R_{\text{PDL}} = 2.021$ consistent at 0.58σ	Numerical result	`PDL_f7_2_verification_v1.py`
Coherent (k^2) scenario rejected at 2.04σ	Numerical result	`PDL_f7_2_verification_v1.py`
^{47}Cr/^{47}V ratio 1.44 (no anomaly)	Observational fact	[1]
OP-E2-PDL: derivation of $\hat{Q}_{E2}$ from C1–C4	Open problem	This document

References

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